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Protein extraction and western blot (mouse tissues) V.1

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Kristin Anderson¹, Breanna Bates¹

¹University of Washington and Fred Hutchinson Cancer Research Center



Kristin Anderson

University of Virginia

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Abstract

This protocol outlines how to extract protein from mouse tissues and running a Western Blot.

In brief:

B6 WT or ID8_{VEGF} mice were dissected and tissues or tumor flash frozen on dry ice and stored at -80°C or immediately lysed with ice-cold RIPA buffer (Thermo Scientific, cat: 89900) containing protease and phosphatase inhibitors (Thermo Scientific, cat: A32965 & 88667) using the gentleMACS Dissociator (Miltenyi). Lysates were centrifuged (12,000 RPM) for 20min at 4°C. Protein concentrations were determined using the Pierce BCA Protein Assay Kit (Thermo Scientific, cat: 23225).

Proteins (20mg/lane) were separated on a mini-protean TGX stain-free gel 4-20% (BioRad; cat: 4568094) and transferred to 0.2mm PVDF membrane (Invitrogen; cat: LC2005). Membranes were blocked for 1h with 5% nonfat dry milk in TBS-T (1xTBS BioRad +0.1% Tween-20) then incubated overnight with 1:100 dilution of MSLN antibody (IBL America; cat: 28127) or 1:1000 Vinculin antibody (R&D systems; cat: MAB6896) in TBS-T. Following incubation with appropriate horseradish peroxidase-conjugated secondary antibodies, proteins were visualized by enhanced chemiluminescence (BioRad; cat: 1705060) on the ChemiDoc XRS system (BioRad).

Materials

MATERIALS

- ✕ Pierce BCA Protein Assay Kit **Thermo Fisher Scientific Catalog #23225**
- ✕ RIPA buffer **Thermo Fisher Scientific Catalog #89900**
- ✕ Protease inhibitor **Thermo Fisher Scientific Catalog #A32965**
- ✕ Phosphatase inhibitor **Thermo Fisher Scientific Catalog #88667**
- ✕ 4-20% Mini-PROTEAN® TGX Stain-Free™ Protein Gels, 10 well, 50 µl **Bio-Rad Laboratories Catalog #4568094**
- ✕ 0.2mm PVDF membrane **Invitrogen - Thermo Fisher Catalog #LC2005**
- ✕ 10x TBS **Bio-Rad Laboratories Catalog #170-6435**
- ✕ C-ERC/Mesothelin (308) Anti-Mouse Rabbit IgG Affinity Purify **IBL America Catalog #28127**
- ✕ Human/Mouse/Rat Vinculin Antibody **Catalog #MAB6896**
- ✕ Clarity™ Western ECL Substrate, 200 ml **Bio-Rad Laboratories Catalog #1705060**
- ✕ gentleMACS M Tubes **Miltenyi Biotec Catalog #130-096-335**

Protein Extraction

- 1 Flash freeze tissues in small chunks on dry ice immediately as dissected store in -80 until ready to use.

Prepare RIPA buffer:

0.876g NaCl

1ml 1M Tris

1ml 10% SDS

100ul Triton X-100

1g Deoxycholate

1ml 0.5M EDTA

Don't forget to add protease inhibitors at the time of sample lysis.

- 2 Add 1 (10x protease) and 1 (10x phosphatase) inhibitor tablet per 10mL of RIPA buffer.
- 3 Add 500 ul ice cold lysis (RIPA) buffer per 10-20mg tissue to M-Tube on ice and add tissue.
- 4 Run program Protein_01 on the gentleMACS Dissociator (Miltenyi).
- 5 Centrifuge M-tube for 1min at 2000xg at 4°C to collect everything and transfer to eppendorf tube on ice.
- 6 Centrifuge eppendorf tubes for 20min at 12,000 rpm at 4°C (tabletop centrifuge).
- 7 Transfer supernatant to new tube on ice and proceed with protein quantification assay.
- 8 Store samples in -80.

Western Blot

- 9 Prepare samples: Add 20ug of protein

Reducing agent (b-mercaptoethanol)
Loading dye (one fourth volume for 4X dye)
Ripa

Total volume (Max) = 20ul for 12-15 well comb and 50ul for 10 well comb.
Heat samples for 90 C for 10 minutes and place on ice. Do not heat molecular marker.

Prepare gel reagents:

10X running Buffer: Tris Base 30.2g, Glycine 144g, SDS 10g

Transfer Buffer: Tris Base 2.42g, Glycine 11.26g, 10% SDS 1ml, Methanol 200ml, Water (make up to 1l)

10X TBS (Cell signaling protocol): Tris Base 48.4g, NaCl 160g, Adjust pH to 7.6 with HCl

TBS-Tween: 1X TBS, 0.1% Tween 20

- 10 To run gel, assemble gel so that wells face the gasket. Fill the gel with running buffer and make sure apparatus is not leaking. Remove combs and flush the wells with 1ml micropipette to remove any gel pieces. Ensure running buffer is full is the inside compartment of the gel box.

For Mesothelin and Vinculin blot, a mini-protean stain-free gel (4-20%) 10 well x 50ul was used.

- 11 Load molecular markers and denatured samples. Connect gel box to power pack. Ensure negative is connected to negative supply of pack (black to black & red to red). Run gel at 60V for 25-30 minutes and then at 100-110V for 60 minutes. You can judge how far the gel should run by the molecular weight markers. Ensure good separation. Make transfer buffer and keep at 4C (as soon as you start running gel).

For Mesothelin and Vinculin blot, 20ug of protein (40ul of sample) was combined with 8ul of Precision Plus Protein Standards, Dual Color.

- 12 Image stainfree gel(s) with chemidoc system after running.
- 13 For each gel to transfer, you need PVDF membrane (0.20um) and 2 pieces of blotting paper cut to the size of the gel (0.5cm bigger on each side).
- 14 Activate PVDF membrane for 1 minute in 100% methanol, wash 3X with water and then soak in transfer buffer on shaker. Immerse the blotting paper and sponge pads in transfer buffer.
- 15 Prepare transfer sandwich. Place white side down, place sponge, place a blotting paper, membrane, gel, blotting paper, sponge and close sandwich.

Try to roll out bubbles after placing the second piece of blotting paper.
Fill the transfer case with transfer buffer, add ice pack & stir bar OR place entire apparatus in ice box to keep transfer apparatus cool. Place sandwich cassettes in apparatus so that black side of cassette faces black side of apparatus. Fill with cool transfer buffer till the top.

- 16 Connect leads. Transfer at 100V for 90 minutes.
- 17 Take down transfer case. Ladders should have transferred to membrane. Mark position of ladders with pen. Always keep protein side up.
- 18 Image blot on Chemidoc for total protein image.
- 19 Block membrane for 60 minutes in 5% non-fat dry milk, TBS-Tween (TBS-T) for 1hr on shaker.
- 20 Wash with TBST to remove milk.
- 21 Add diluted primary antibody (Can dilute in 5% BSA in TBS-T) overnight at 4C on a shaker.

Mesothelin primary antibody: ~40-70kDa diluted 1:75 = 133ul in 10ml TBS-T (Rabbit, from IBL 28127)
Vinculin primary antibody: ~125KDa diluted 1:1000 = [1ug/mL] (Monoclonal Mouse, from R&D systems)
- 22 Wash 5 time for 5 minutes with TBS-T.
- 23 Add diluted secondary antibody conjugated to HRP for 1 hour at room temperature.

For Mesothelin and Vinculin blot:
Goat Anti-Rabbit IgG-HRP (Thermo) = 1:1500
Goat Anti-Mouse IgG-HRP (Santa Cruz) = 1:3000
- 24 Wash 5 time for 5 minutes with TBS-T.
- 25 Develop with ECL solution according protocol on specific product. Ensure ECL solution covers membrane for approximately 5 minutes.



- 26 Let excess ECL solution drip off on Kimwipe.
- 27 Expose to film/develop blots in dark room and on ChemiDoc System.
- 28 Can store membrane at 4C in saran wrap.